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Reproducibility

Data Engineering and Analytics

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Reproducibility

- A paper reports 92% accuracy on a classification task.
- You download the code and data, run the experiment, and get 86%.
- What could explain the difference?

Potential causes:

- Dataset version or train/test split?
- Preprocessing or hyperparameters?
- Random seed, dependencies, or hardware?

Why Reproducibility Matters

- Reliability: detect mistakes and check the robustness of results.
- Transparency: make assumptions, methods, and decisions inspectable.
- Reuse: build on prior work instead of reverse-engineering it.
- Collaboration: help colleagues—and your future self—rerun and extend analyses.
- Good scientific practice: make claims verifiable and contribute to cumulative research.
- A result is more useful when others can verify how it was produced.

Evidence: Reproducibility in Recommender Systems (2019)

- Ferrari Dacrema et al. (2019) examine experimental practices in neural recommender systems.
- Their analysis highlights weaknesses in comparisons with baselines and difficulties in reproducing reported improvements.
- Take-away: Releasing code and data is important, but does not guarantee sound experimental evaluation.

Conference	Rep. Setup ratio	Reproducible Setup	Non-Reproducible Setup
KDD	3/4 (75%)	[32], [37], [73]	[67]
IJCAI	5/7 (71%)	[29], [80], [79], [13], [77]	[51], [76]
WWW	2/4 (50%)	[30], [38]	[66], [22]
SIGIR	1/3 (30%)	[21]	[48], [12]
RecSys	1/7 (14%)	[81]	[65], [8], [60], [68], [34], [71]
WSDM	0/1 (0%)		[75]
Total	12/26 (46%)		

Ferrari Dacrema, M., Cremonesi, P., & Jannach, D. (2019). *Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches.* RecSys '19, 101–109. DOI

Evidence: Reproducing AI Research (2025)

- Gundersen et al. attempted to reproduce 30 highly cited AI studies.
- 8 studies were excluded from the final assessment because the required data or hardware were practically inaccessible.
- Of the remaining 22 studies, 6 were fully and 5 partially reproduced (50% at least partially reproduced).
- Among studies sharing both code and data, 86% were fully or partially reproduced; among those sharing only data, the figure was 33%.
- Take-away: Sharing code and data is strongly associated with successful reproduction; good data documentation also matters.

Gundersen, O. E., Cappelen, O., Mølnå, M., & Nilsen, N. G. (2025). *The Unreasonable Effectiveness of Open Science in AI: A Replication Study.* AAAI 39(25), 26211–26219. DOI

Evidence: Artifact Consistency in RecSys (2026)

- Ferrari Dacrema, Benigni & Ferro analyzed 10 graph-based recommender-system papers, mostly from SIGIR 2022.
- They report problems including erroneous data splits and information leakage.
- Source code and data artifacts were often inconsistent with descriptions in the papers.
- Comparisons sometimes favored complex but weak baselines over simpler, stronger alternatives.
- Take-away: Available artifacts are not sufficient: data splits, baselines, and consistency between paper and implementation must also be checked.

Ferrari Dacrema, M., Benigni, M., & Ferro, N. (2026). *Reproducibility and Artifact Consistency of the SIGIR 2022 Recommender Systems Papers Based on Message Passing.* ACM Transactions on Information Systems, 44(1), Article 23. DOI

Evidence: Reproducibility Research in RecSys (2026)

- Said & Bellogín survey the 51 papers accepted to the ACM RecSys Reproducibility Track (2020–2025); 50 are included in their detailed RecSys analysis.
- The track has expanded from reproduction studies to include benchmarking, resources, and methodological contributions.
- Studies frequently reuse a limited set of datasets, algorithms, and evaluation protocols.
- Many studies extend earlier experiments rather than reproduce every original condition exactly.
- Take-away: Reproducibility research improves transparency, but its methodological diversity remains limited.

Said, A., & Bellogín, A. (2026). *Reproducibility in Recommender Systems: A Survey.* ACM Transactions on Recommender Systems. DOI

Definitions

- We use the Association for Computing Machinery (ACM)’s definitions in the following.
- ACM definitions are aligned with the NISO RP-31-2021, Reproducibility Badging and Definitions document; approved by the (US) National Information Standards Organization (NISO).

- ACM uses these definitions for badges annotating papers: 

Note that in previous versions, ACM has used “swapped” definitions reproducibility and replication. Reasoning for the change: “As a result of discussions with the National Information Standards Organization (NISO), it was recommended that ACM harmonize its terminology and definitions with those used in the broader scientific research community, and ACM agreed with NISO’s recommendation to swap the terms”reproducibility” and “replication” with the existing definitions used by ACM as part of its artifact review and badging initiative.” (ACM artifact review and badging)

Overview of Terms

Term	Who?	Experimental setup?
Repeatability	Same team	Same
Reproducibility	Different team	Same (author’s artifacts)
Replicability	Different team	Independently developed

Scenario: Can someone else rerun your published training pipeline? That tests reproducibility.

Reproducibility

The measurement can be obtained with stated precision by a different team using the same measurement procedure, the same measuring system, under the same operating conditions, in the same or a different location on multiple trials. For computational experiments, this means that an independent group can obtain the same result using the author's own artifacts.

(Different team, same experimental setup)

Repeatability

The measurement can be obtained with stated precision by the same team using the same measurement procedure, the same measuring system, under the same operating conditions, in the same location on multiple trials. For computational experiments, this means that a researcher can reliably repeat her own computation.

(Same team, same experimental setup)

Replicability

The measurement can be obtained with stated precision by a different team, a different measuring system, in a different location on multiple trials. For computational experiments, this means that an independent group can obtain the same result using artifacts which they develop completely independently.

(Different team, different experimental setup)

Degrees of Reproducibility

Reproducibility is not a binary property; it exists in degrees depending on how much of the experimental setup is shared and how much is independently verified.

Methods reproducibility: The ability to implement, as exactly as possible, the experimental and computational procedures, with the same data and tools, to obtain the same results.

Results reproducibility: The production of corroborating results in a new study, having used the same experimental methods.

Inferential reproducibility: The drawing of qualitatively similar conclusions from either an independent replication of a study or a reanalysis of the original study.

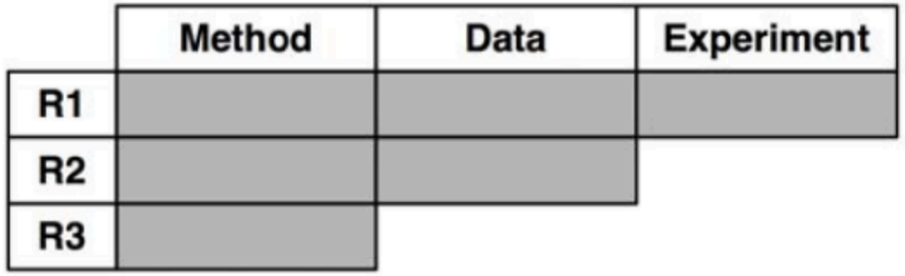


Figure taken from Gundersen, O. E., & Kjensmo, S. (2018). State of the Art: Reproducibility in Artificial Intelligence. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1).

Goodman SN, Fanelli D, Ioannidis JP. What does research reproducibility mean? Sci Transl Med. 2016 Jun 1;8(341):341ps12. doi: 10.1126/scitranslmed.aaf5027. PMID: 27252173.

Degrees and Requirements

Question: What can an independent team verify with the available artifacts?

- Description: Can I understand what was done?
- Artifacts: Can I inspect the code, data, and configuration?
- Execution: Can I run the workflow and obtain the stated results?
- Independent verification: Can others check the conclusions?

Document the Entire Experimental Pipeline

Data → preprocessing → training/analysis → evaluation → results

At each step, record:

- Inputs: versions, sources, and splits.
- Instructions: code, parameters, and dependencies.
- Outputs: models, intermediate results, and metrics.

Goal: another person can follow the same path without guessing.

Example: the missing experiment specification

```yaml
# experiment.yaml
seed: 42
dataset: "customer_churn"
data_version: "v1.2"
split:
  strategy: stratified
  test_size: 0.2
model:
  type: random_forest
  n_estimators: 200
evaluation:
  metric: accuracy
```

A configuration file makes key choices explicit and reviewable.

A reproducible project has an entry point

```
project/
├── README.md
├── requirements.txt
├── experiment.yaml
├── src/
│   ├── prepare_data.py
│   ├── train.py
│   └── evaluate.py
├── data/README.md
└── results/README.md
```

The README should answer: How do I set up, run, and check this project?

Before Publishing: Reproducibility Checklist

Area	Questions to ask
Data	Provenance, version, preprocessing, splits?
Code	Source, dependencies, run instructions?
Experiment	Seeds, configuration, hyperparameters, environment?
Results	Metrics, baselines, intermediate outputs?

For fuller research checklists see Pineau et al. (2021), JMLR 22(164), 1–20, and the NeurIPS or AAAI paper checklist.

Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., ... & Larochelle, H. (2021). Improving reproducibility in machine learning research (a report from the neurips 2019 reproducibility program). Journal of machine learning research, 22(164), 1-20.

Reproducibility: What to Report

- State underlying assumptions: The assumptions the model relies upon (e.g., data properties) should be clearly stated and, when necessary, verified to be true for the evaluation setup at hand.
- Be clear about data pre-processing: All pre-processing steps should be described and the removal of data should be clearly motivated.
- Report hyperparameter optimization details: The article should either explicitly mention or provide a reference to the range and distribution of each hyperparameter, and should report the method that was used to select them. This includes the number of epochs the model was trained for and early-stopping criteria.
- These hints and checklists stem from academia, but most points are equally relevant for an industry-setting as well as master theses.

Maurizio Ferrari Dacrema, Paolo Cremonesi, and Dietmar Jannach. 2019. Are we really making much progress? A worrying analysis of recent neural recommendation approaches. In Proceedings of the 13th ACM Conference on Recommender Systems (RecSys '19). Association for Computing Machinery, New York, NY, USA, 101–109. DOI

Reproducibility: Implementation

- Publish the source code of all models, including the baselines.
- Provide the datasets: The used datasets, or pointers to them, should be published. If possible, the train and test splits should be published as well.
- Make the reproduction easy for others: Authors should specify all software requirements and dependencies of their software, including the specific version. Provide installation scripts and machine-readable dependency descriptions. It is also desirable to provide scripts that execute the entire experimental pipeline, from pre-processing, to training, evaluation and collection of metrics.
- Use persistent repositories: Institutional repositories or DOIs should be preferred to personal webpages since the latter can have a much shorter lifespan making previously published links unavailable.

Maurizio Ferrari Dacrema, Paolo Cremonesi, and Dietmar Jannach. 2019. Are we really making much progress? A worrying analysis of recent neural recommendation approaches. In Proceedings of the 13th ACM Conference on Recommender Systems (RecSys '19). Association for Computing Machinery, New York, NY, USA, 101–109. DOI

Reproducibility: Implementation

- Releasing research code checklist and README template
- AAAI 2027 Reproducibility Checklist (part of full author kit)
- NeurIPS Paper Checklist Guidelines
- NeurIPS Code and Data Submission Guidelines

Open Research Data

- FAIR principles (Findable, Accessible, Interoperable, Reusable)
- Metadata
- Description
- Also describe potential flaws, preprocessing.
- Datasheets for datasets for documentation of datasets.
  - Where did the data come from, and why was it collected?
  - What does each field mean?
  - What preprocessing or filtering was performed?
  - What biases, gaps, or limitations are known?
  - Who may access and reuse it, and under which conditions?

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. Commun. ACM 64, 12 (December 2021), 86–92. DOI

FAIR Principles

- Reuse of (scholarly) data.
- Guideline for enhancing reusability of data.
- Optimized for automatically finding and using the data.
- → use a suitable repository, such as Zenodo, when the data can be shared.
- FAIR does not necessarily mean open: access restrictions can be compatible with FAIR when access conditions are clearly specified.

Wilkinson, M. D. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci. Data 3:160018, doi: 10.1038/sdata.2016.18 (2016). Zenodo FAIR principles

Tooling

Tools

- In the following, we present a few (Python-based) tools for increasing reproducibility of projects and analyses.
- In addition to existing tools:
  - Git
  - Static code analysis
  - Pre-commit hooks
  - Virtual environments
  - Pipelines
  - Python modules
- Compilation of MLOps tools

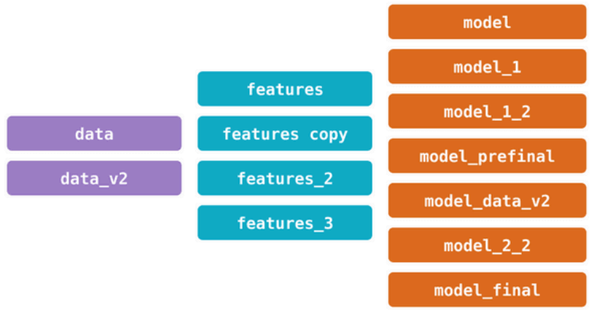
Choose tools by the problem they solve

Problem | Possible tool or practice |
——————————— | ——————————————— |
Code changes unexpectedly | Git |
Dataset or model files change | DVC, lakeFS, Delta Lake |
Environment differs | Virtual environments, pinned dependencies |
Runs and metrics get mixed up | MLflow, Weights & Biases |
Steps are executed manually | Automated pipelines |

Tools support reproducibility; they do not replace documentation.

Data Versioning

- Data versioning is crucial for reproducibility, especially when working with large datasets that evolve over time.
- Not only version code, but also data (raw data, preprocessed data, intermediate results, and final outputs).
- Avoiding situations like



- Example tools for data versioning include:
  - DVC
  - Delta Lake
  - LakeFS

Figure taken from the DVC documentation

Data Version Control (DVC)

- Key idea: versioning of data and machine learning models.
- Track which data and model artifacts belong to a particular experiment.
- Git-like workflow for versioning large data and model artifacts.
- Keep large artifacts in suitable remote storage rather than normal Git history.
- No need to adapt e.g., file names.
- Allows to store data and models to cloud storage (e.g., AWS) or via SSH/HTTP/WebDav etc.
- Define reproducible pipeline stages and dependencies.

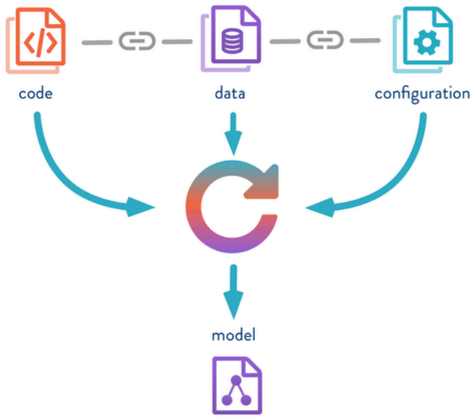


Figure taken from the DVC documentation

Data Version Control (DVC)

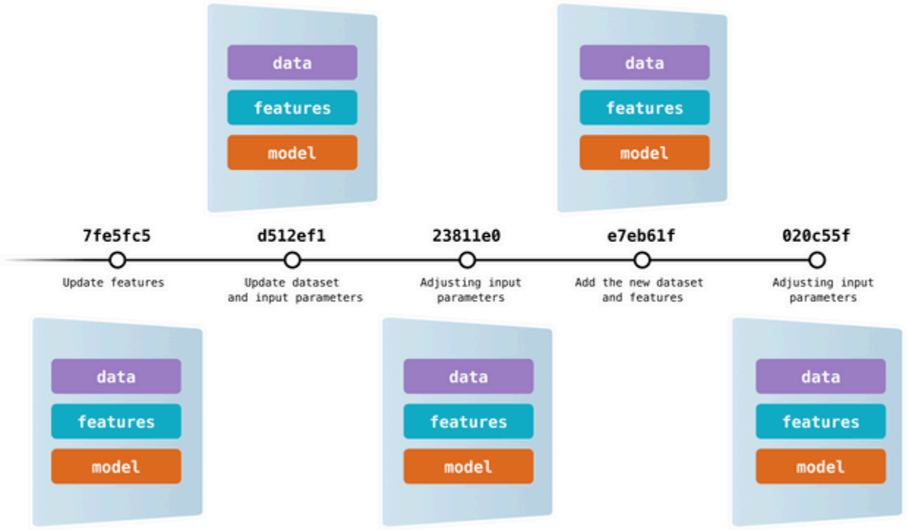


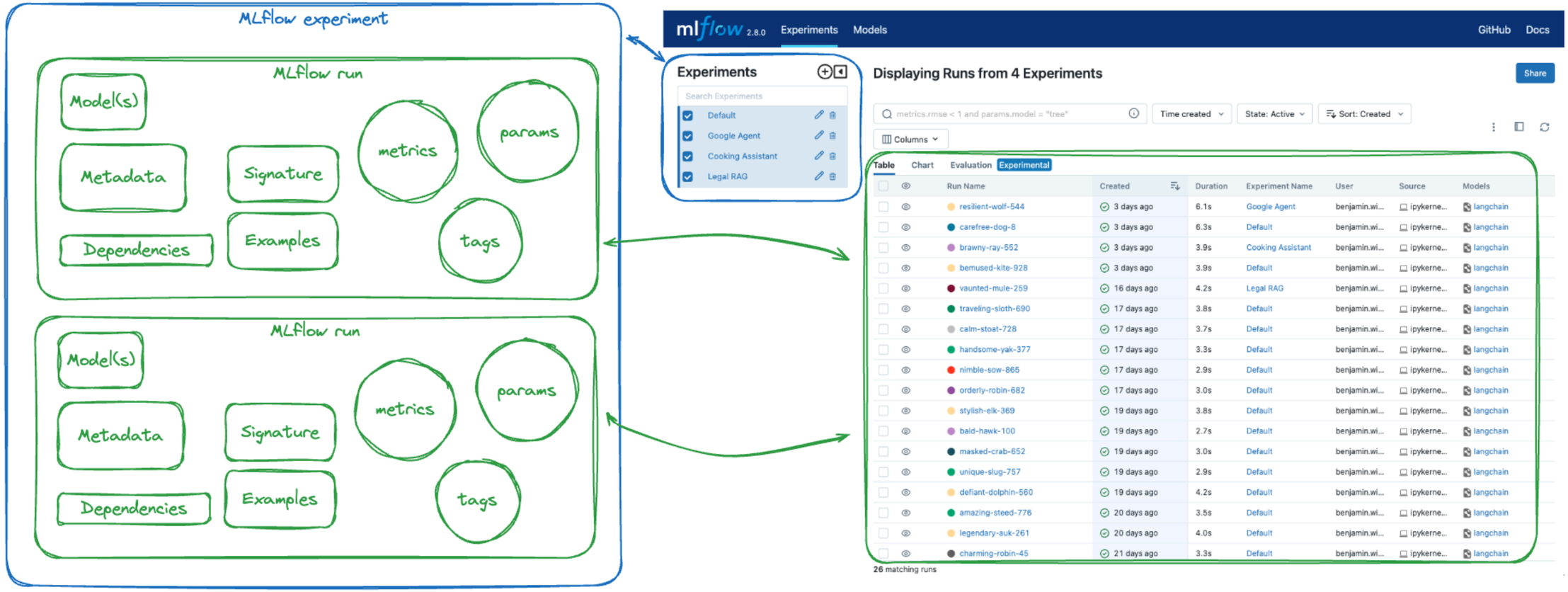
Figure taken from the DVC documentation

Model Versioning and Training

For each run, capture:

- Code and data versions.
- Parameters and random seed.
- Environment and execution details.
- Metrics, outputs, and model artifacts.
- Example tools for capturing this information include:
  - Weights and Biases
  - MLflow

Model Versioning and Training



Further Tools

- Further MLOps tools
- Includes tools for model lifecycle, model serving, hyperparameter tuning, and workflow management.

Bottom Line Question: Can another team rerun your project?

Before handing off an analysis, check:

- Can they find and access the required data?
- Can they recreate the software environment?
- Can they execute the entire workflow?
- Can they compare their results with documented expectations?
- Can they understand deviations and limitations?